

Distributed INS-1 signaling through the candidate noncanonical receptor NTR-1 tunes the ASER-AIY axis to drive high-salt ethanol avoidance in *Caenorhabditis elegans*

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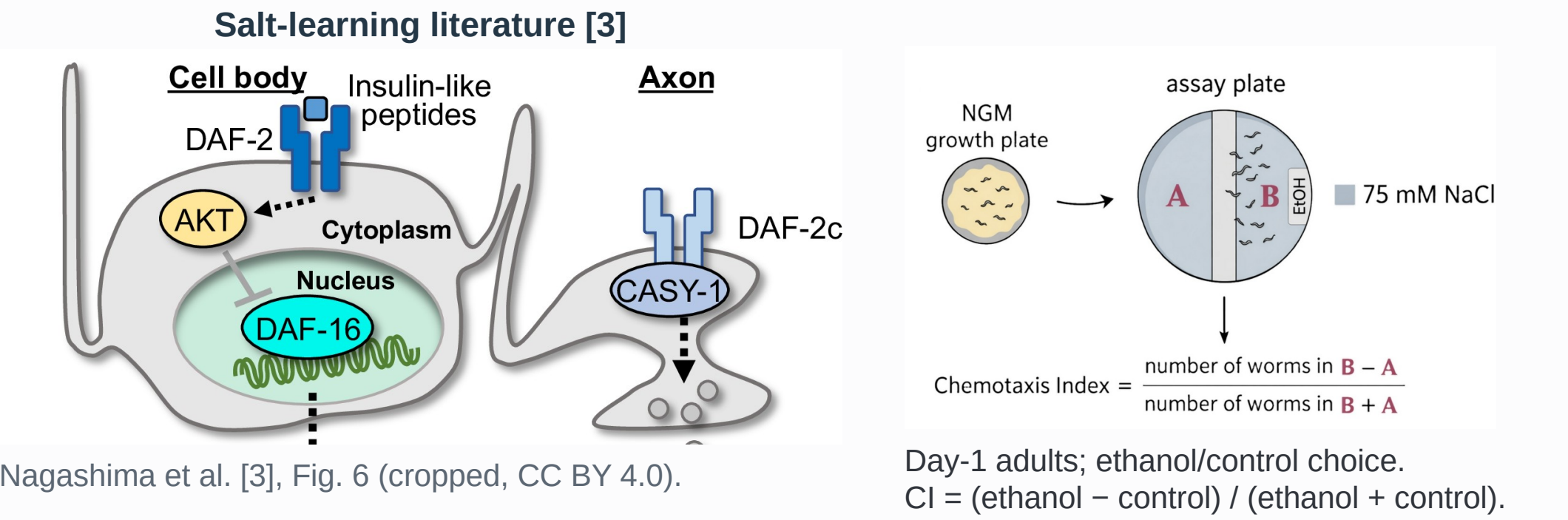
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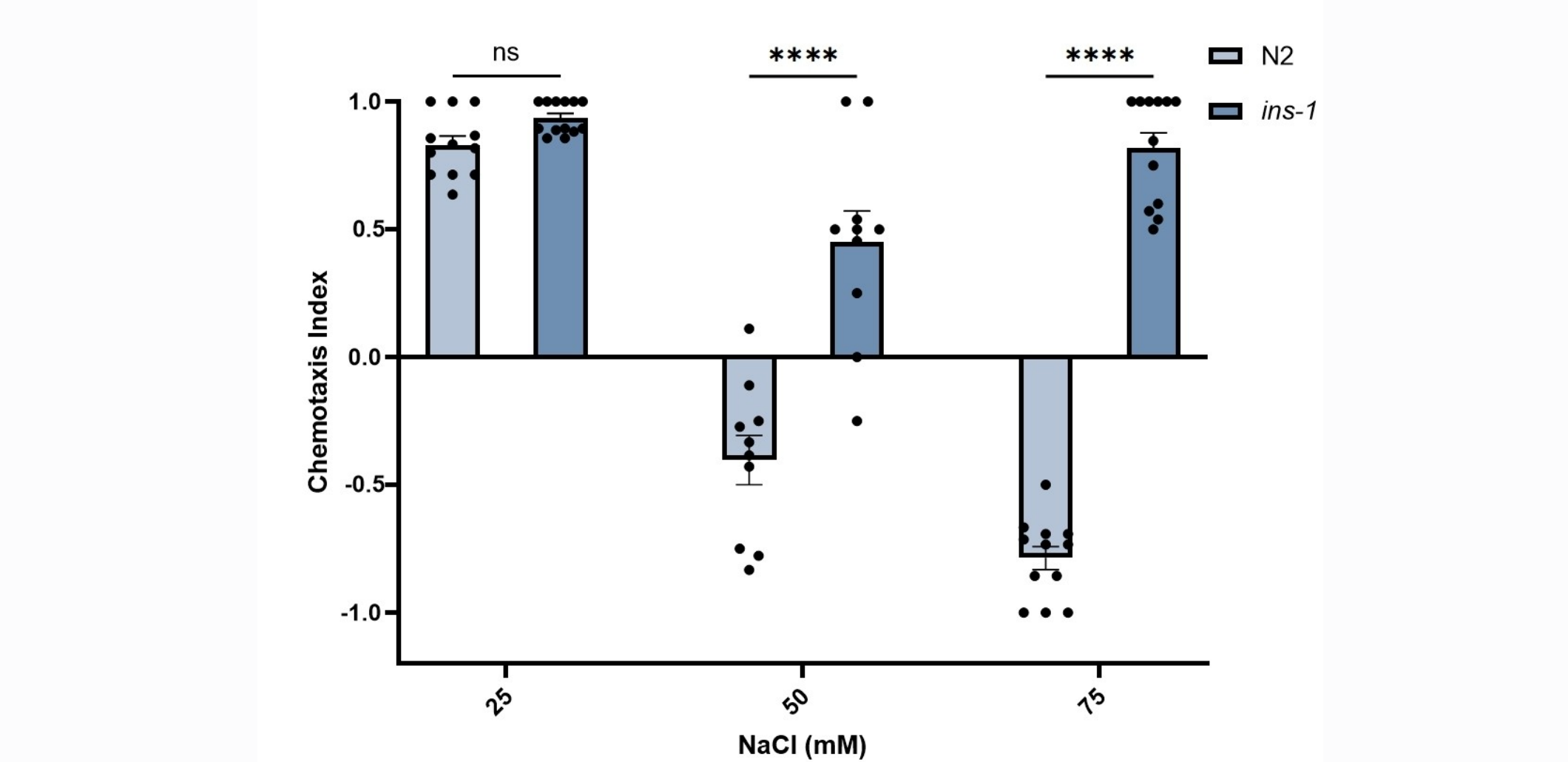
BACKGROUND & BEHAVIOR

Background and assay

ASER and ASER encode salt increases and decreases [1]. INS-1 modulates DAF-2/PI3K signaling in ASER [2,3], while nematocin–NTR-1 signaling supports gustatory plasticity [4]. We test how INS-1 tunes ethanol valence under high salt.

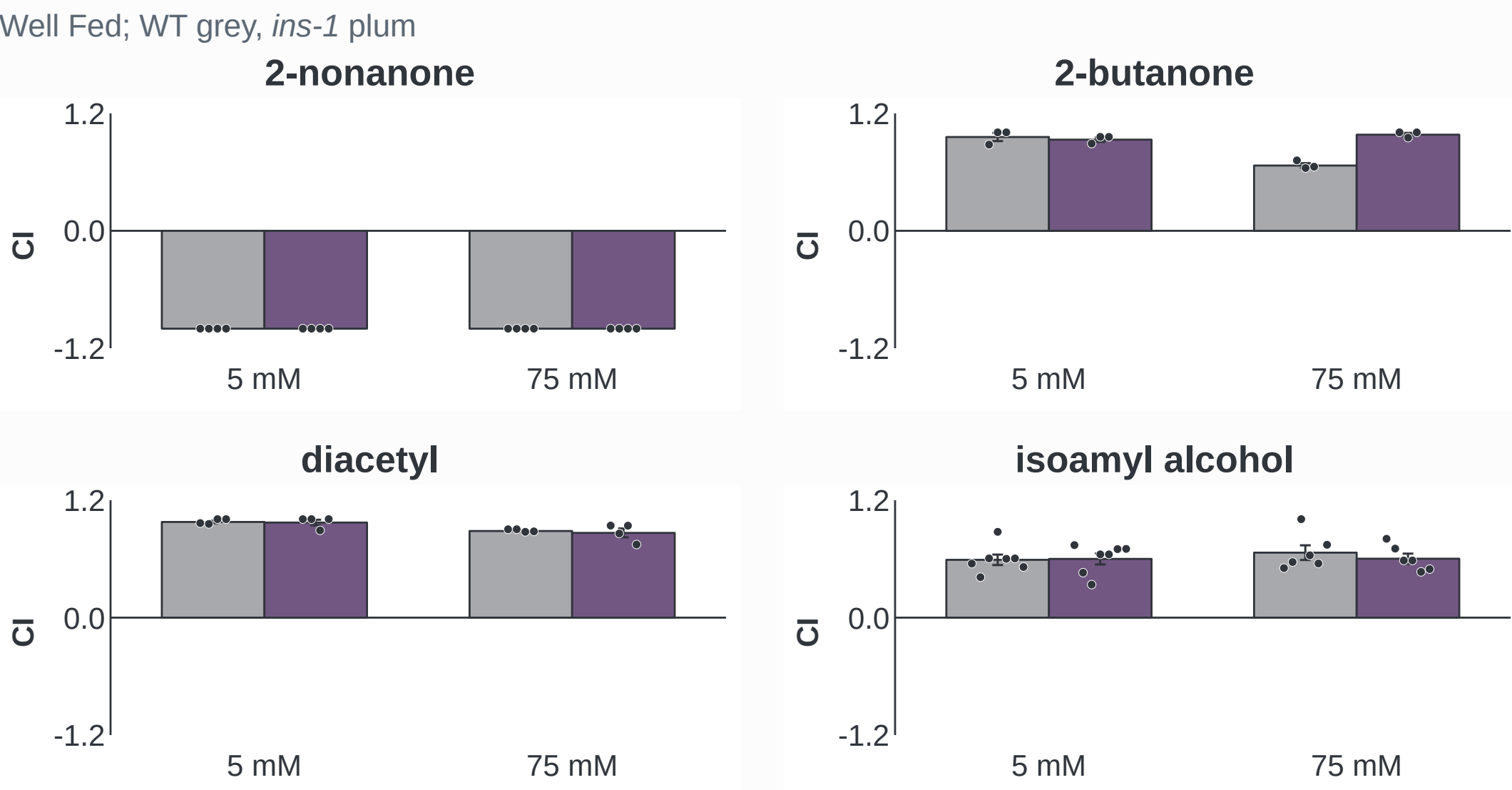


INS-1 loss reverses high-salt avoidance



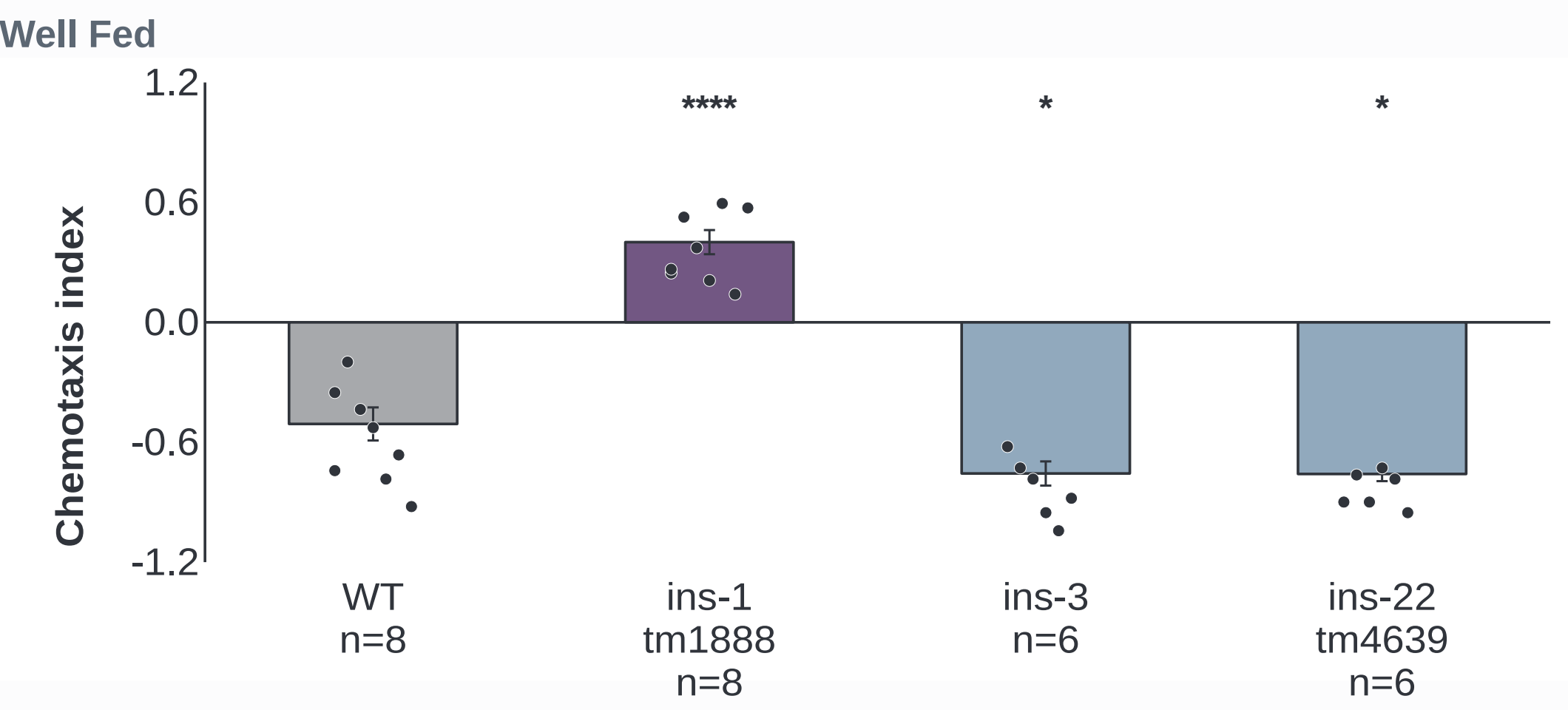
Mean $\pm$ SEM; $n > 8$ assays. Positive CI, attraction; negative CI, avoidance. ns and **** $P < 0.0001$ are source reported.

Chemotaxis to other odorants



Well Fed. Culture: 50 mM NaCl; assay NaCl shown on each axis. Mean $\pm$ SEM; $n = 3-7$ plates/group. Mean avoidance of 2-nonanone and attraction to the other three odorants persisted in these tested conditions.

Insulin-like peptide mutant screen



75 mM NaCl. Mean $\pm$ SEM; symbols vs WT (Welch–Holm, 3 tests). Only *ins-1* reversed mean valence.

CELL-SPECIFIC INS-1 PERTURBATION

Cell-specific deletion and rescue on one shared CI axis

Well Fed. Genotype colors apply to native charts. Mean $\pm$ SEM; each behavioral point is an independent plate. ASER dose is encoded by shade.

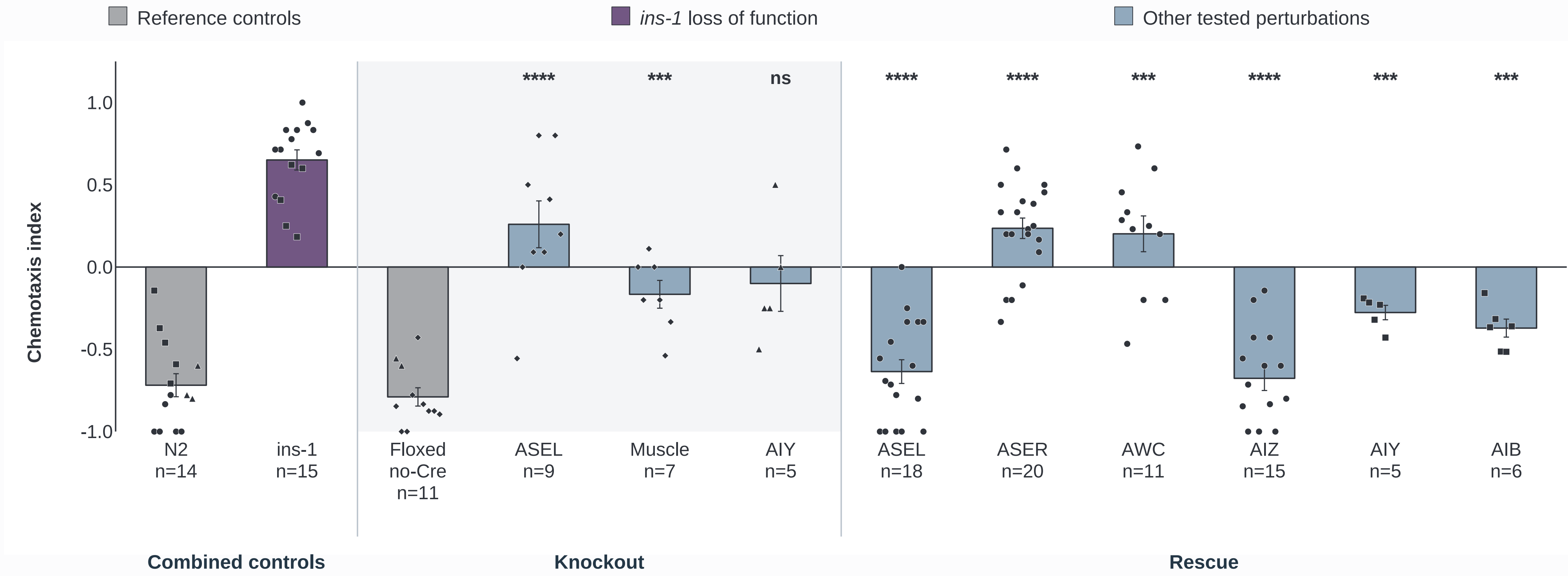
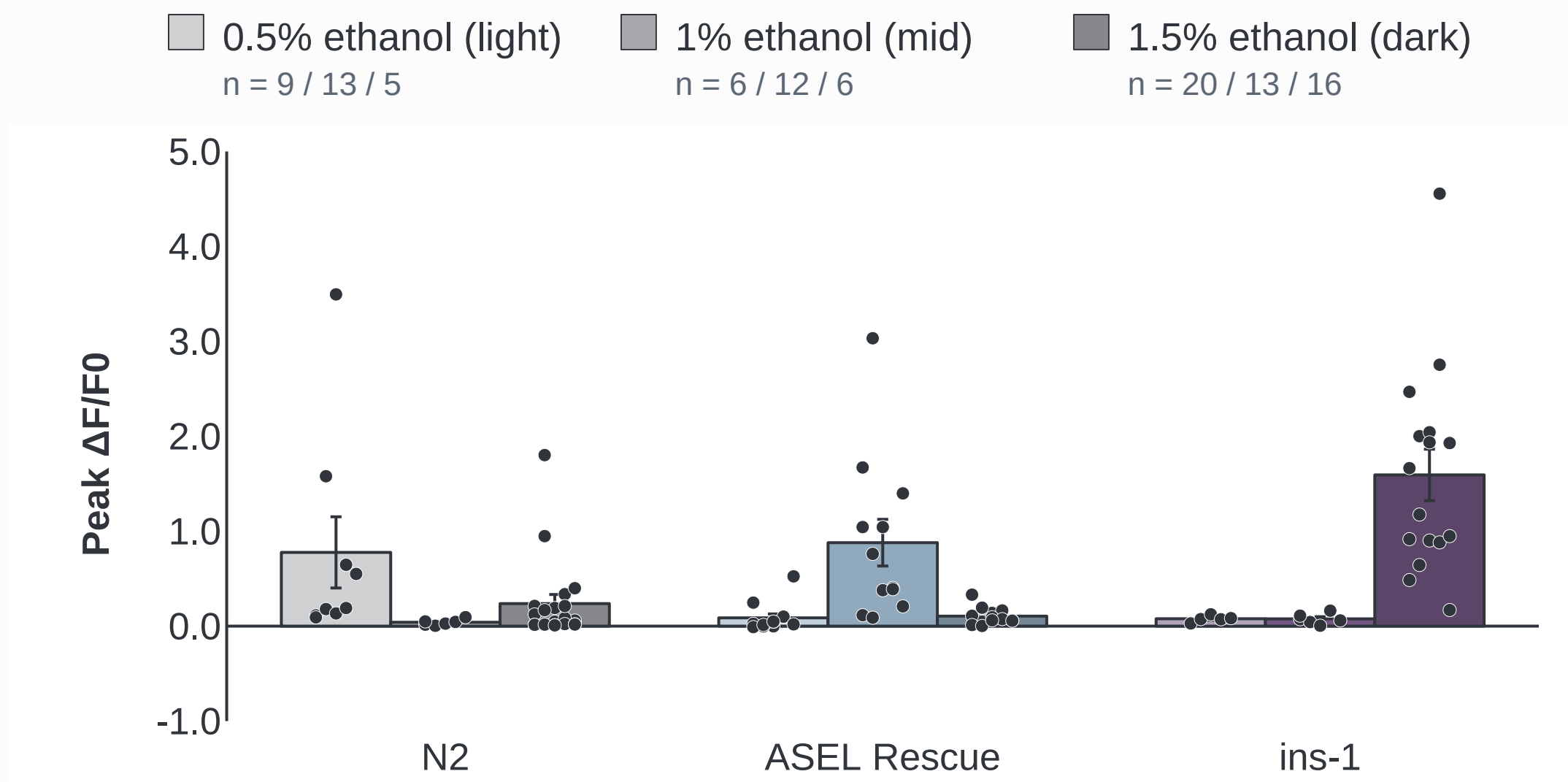


Plate sources: circles, ASER/ASER/AWC/AIZ rescue; diamonds, ASER/muscle knockout; triangles, AIY knockout; squares, AIY/AIB rescue.

Primary symbols: knockout vs source-matched floxed no-Cre (5 tests); rescue vs source-matched *ins-1* (8 tests). Two-sided Welch–Holm. Combined control bars are descriptive. AIY/AIB rescue uses *txx-3p* / *npr-9p*.

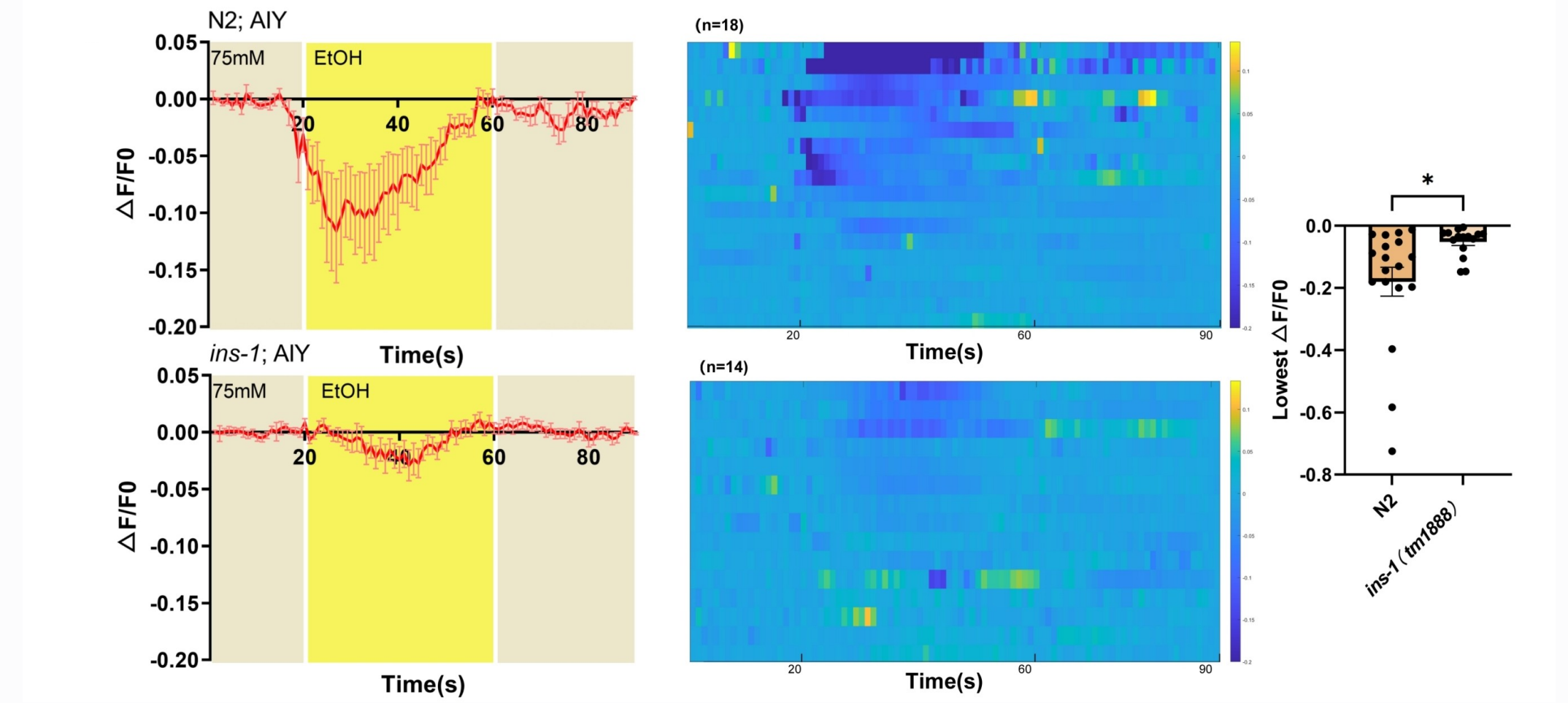
ASER rescue shifts ASER sensitivity



Adjusted P (Welch + Holm, 9 tests). n order: N2 / rescue / *ins-1*.

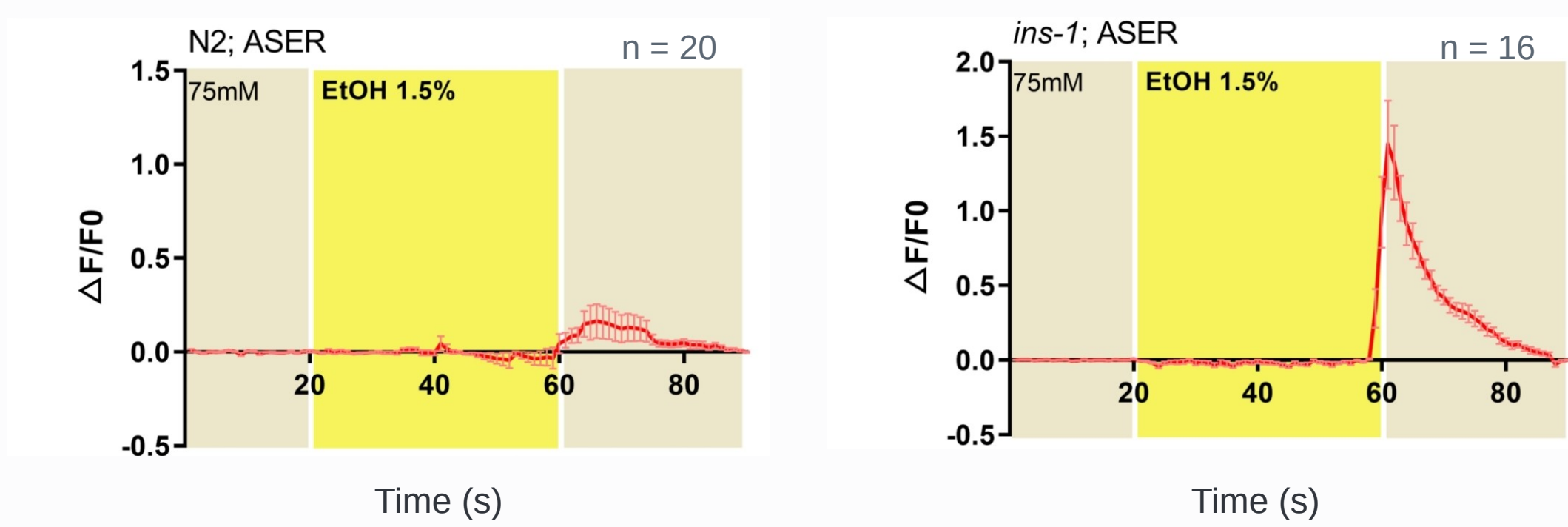
EthOH	N2 vs rescue	N2 vs <i>ins-1</i>	rescue vs <i>ins-1</i>
0.5%	0.4970	0.4970	0.8424
1%	0.0407	0.5885	0.0454
1.5%	0.5885	0.00125	0.000557

AIY negative responses are reduced

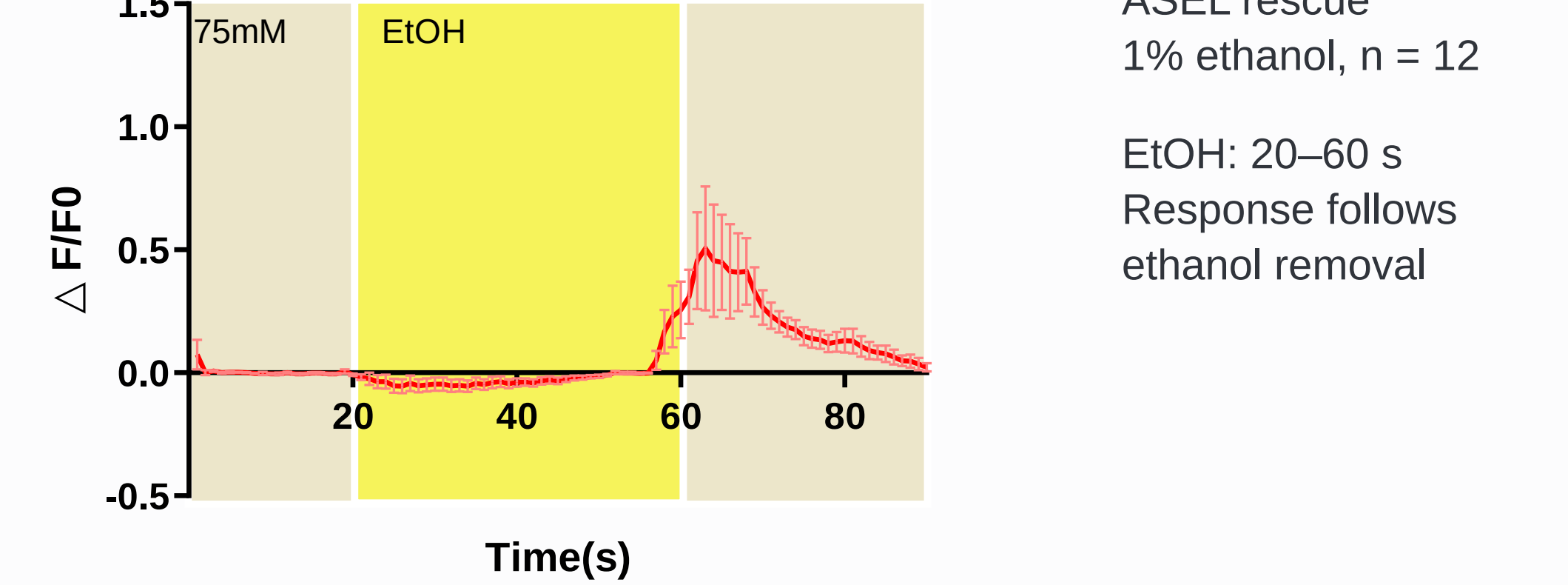


AIY original cohort: N2 $n = 18$; *ins-1* $n = 14$; * $P < 0.05$, source reported.

ASER responses to ethanol removal

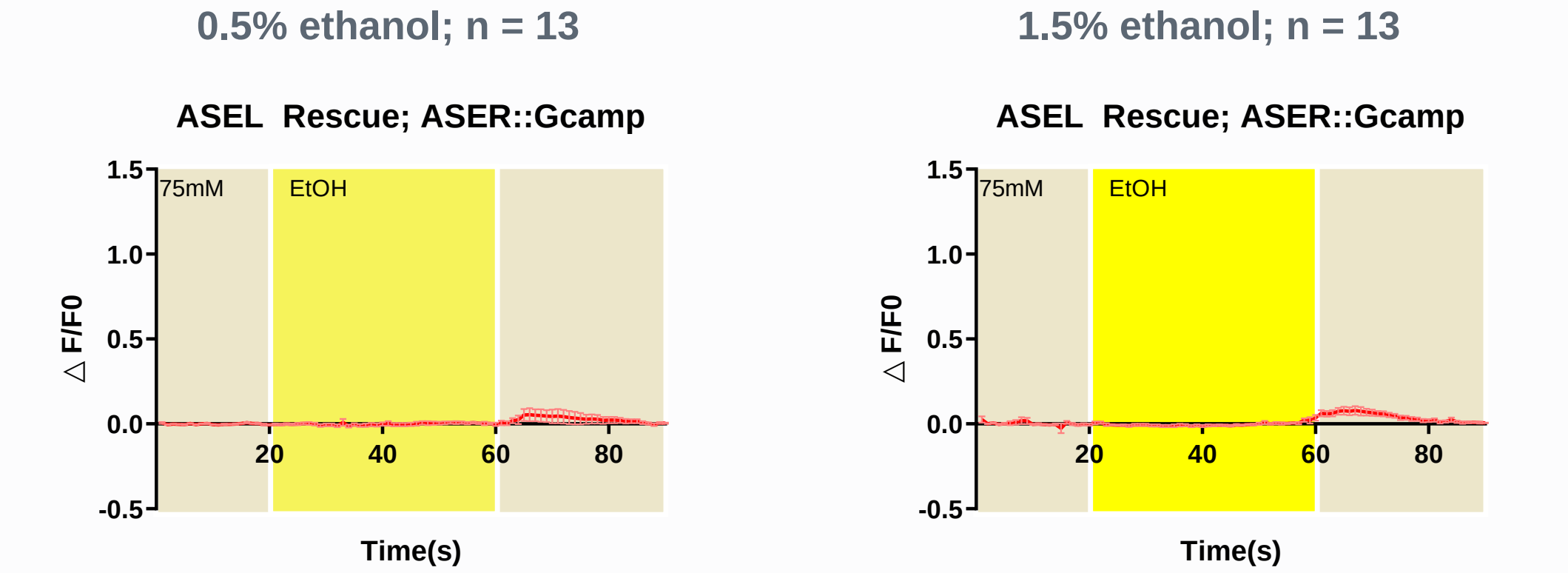


ASER Rescue; ASER::Gcamp



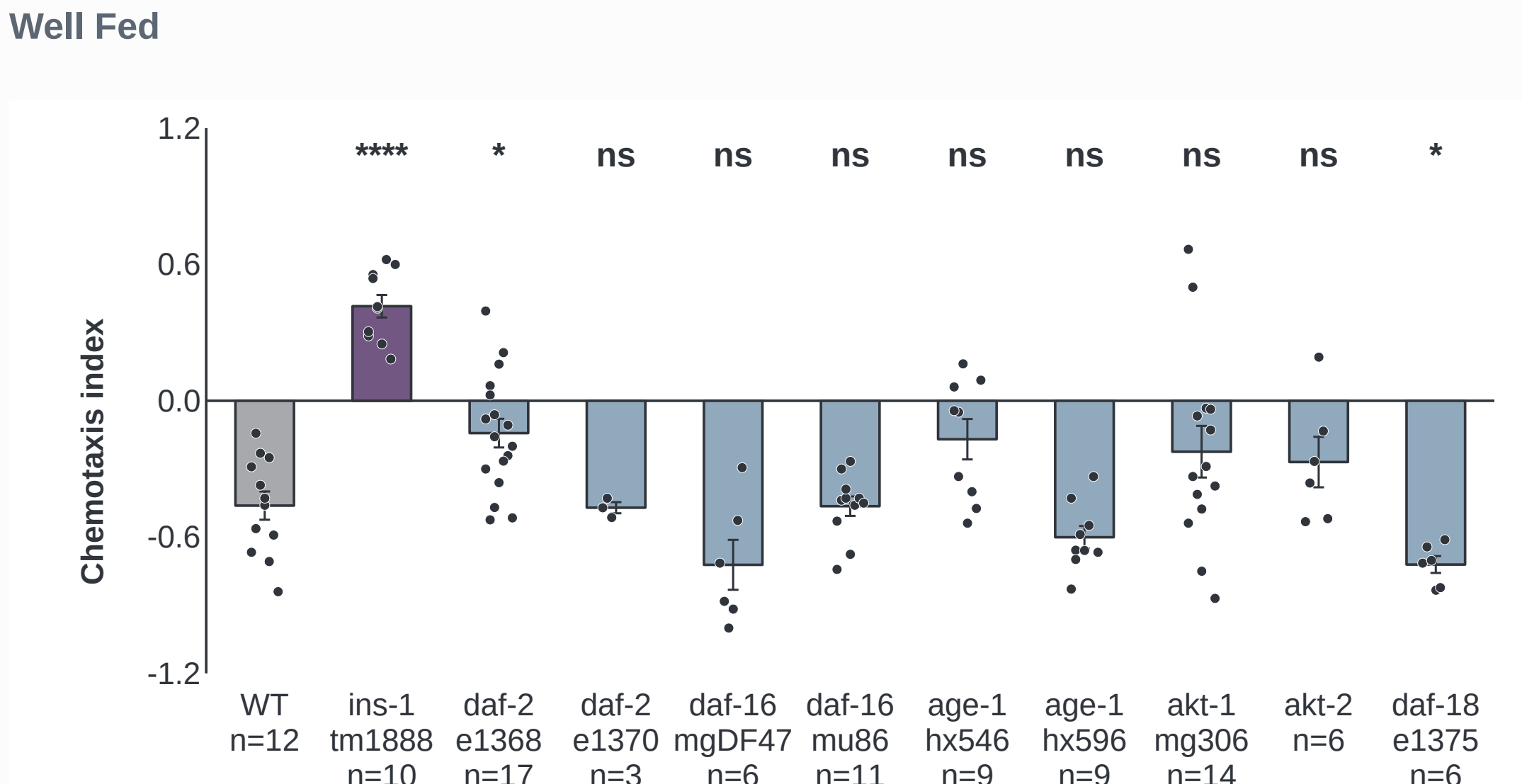
At 1.5%, ASER responds more strongly in *ins-1* than N2. ASER rescue responds most strongly to 1% among the tested concentrations; no continuous threshold is inferred.

ASER rescue: flanking ethanol doses



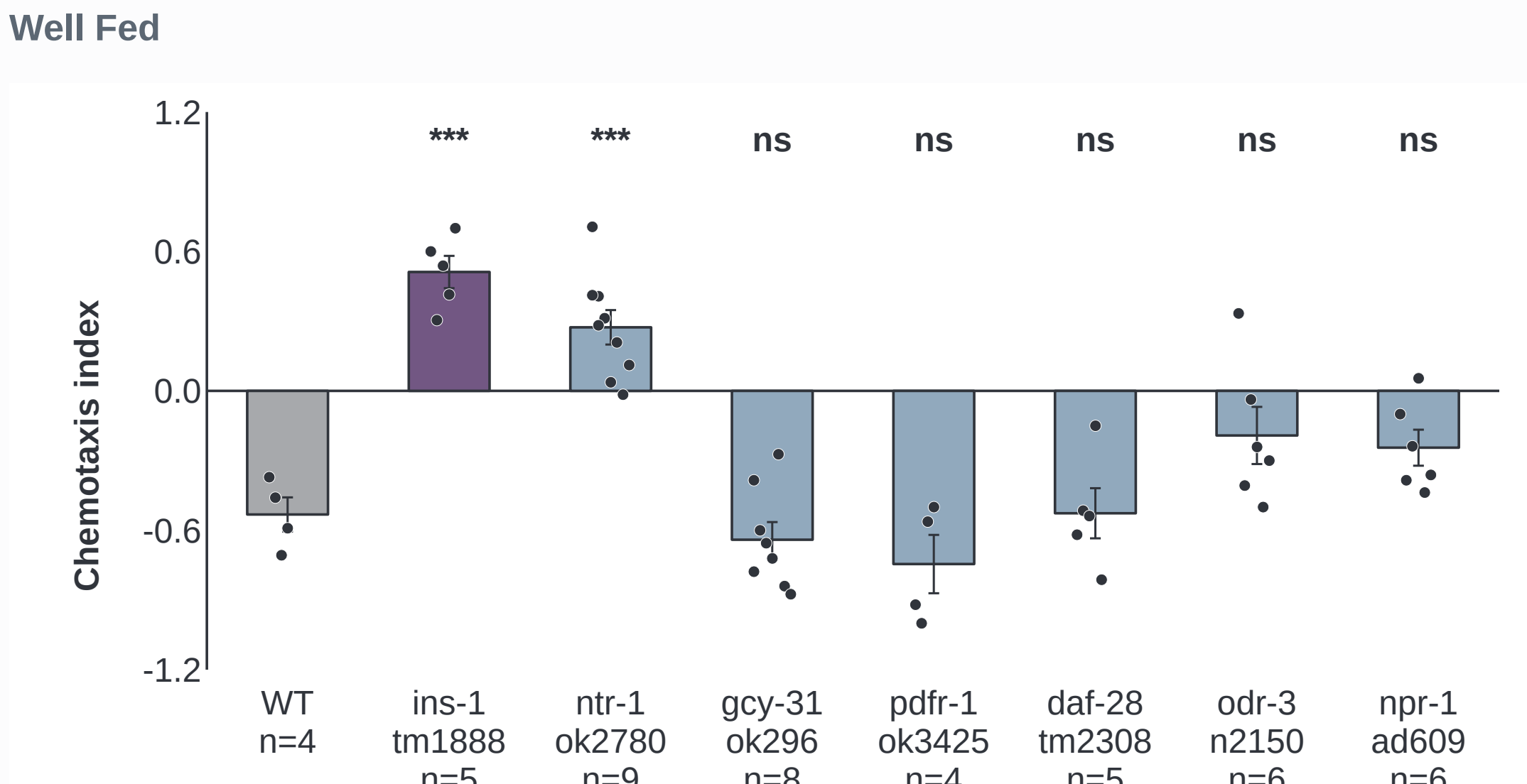
MUTANT SCREENS & MODEL

Canonical insulin-signaling mutant screen



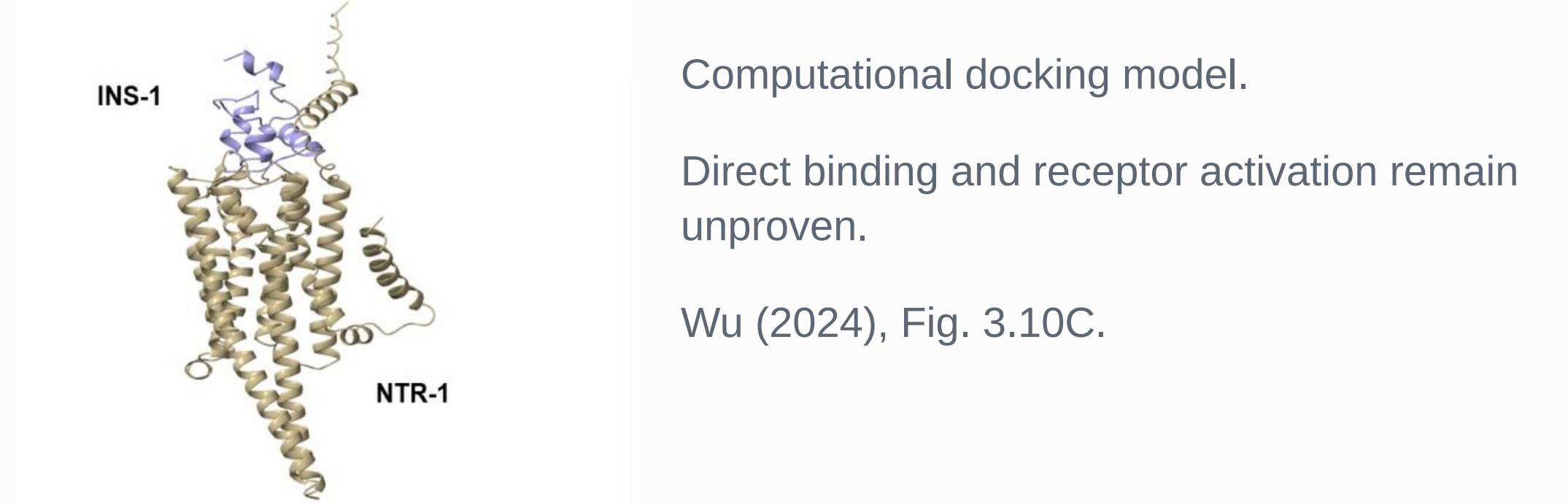
75 mM NaCl. Mean $\pm$ SEM; symbols vs WT (Welch–Holm, 10 tests). Only *ins-1* reversed mean valence among these tested IIS mutants.

Candidate receptor and signaling screen

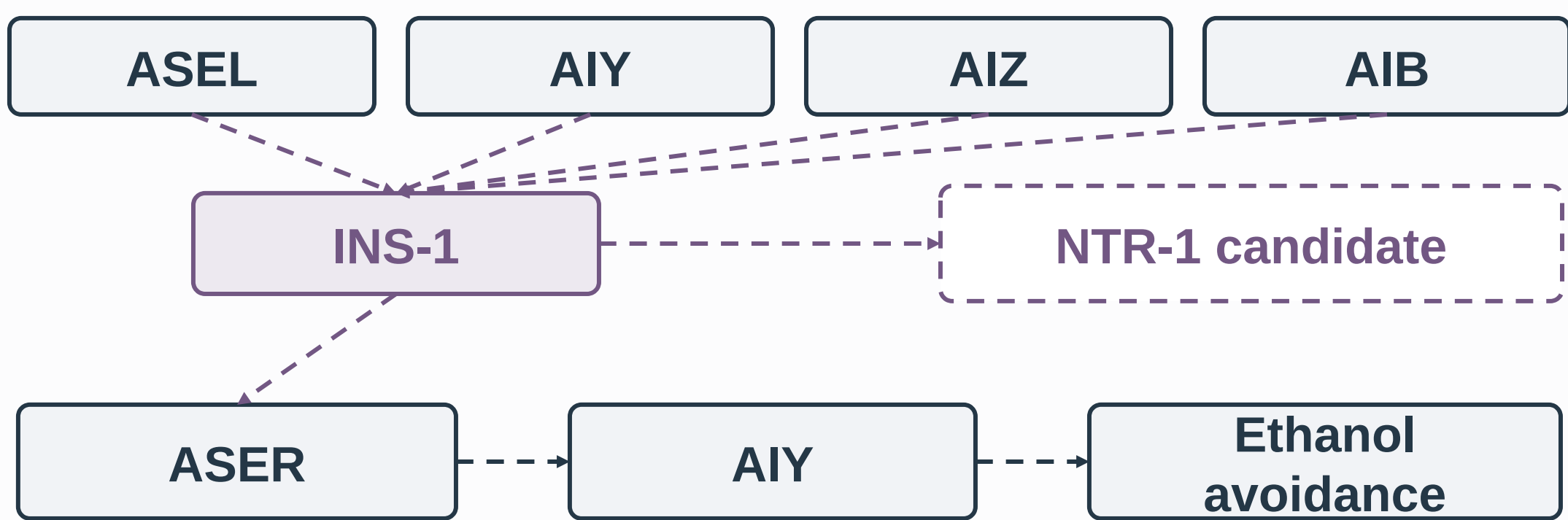


75 mM NaCl. Mean $\pm$ SEM; all 47 selected plate values. Symbols compare each mutant with WT (Welch–Holm, 7 tests). NTR-1 remains a candidate receptor.

Predicted INS-1–NTR-1 complex



Working circuit model



INS-1 tunes ASER–AIY sensitivity and ethanol valence.